

Non Homogeneous Linear Recurrence Relation with constant coefficients

Suppose the k^{th} order is

$$C_0 a_n + C_1 a_{n-1} + C_2 a_{n-2} + \dots + C_k a_{n-k} = f(r) \quad \text{--- (1)}$$

The solution of eqⁿ 1 is

$$a_n = a_n^{(h)} + a_n^{(p)} \quad \text{--- (2)}$$

$a_n^{(h)} \rightarrow$ Homogeneous solution

$a_n^{(p)} \rightarrow$ particular solution

$\rightarrow$ Solution of Homogeneous Linear Recurrence Relation when $f(n) = 0$

$$C_0 a_n + C_1 a_{n-1} + C_2 a_{n-2} + \dots + C_k a_{n-k} = 0$$

$\rightarrow$ Method of finding particular solution when $f(n) \neq 0$

To find the PS (particular solⁿ) of RL (Recurrence Relation), Trial solution on the basis nature of $f(r)$ will be considered

Case I

Case I

When $f(n) = b^n$ & b is not a root of characteristic equation

Suppose trial solⁿ is $a_n = A \cdot b^n$

↳ constant value for which we have to find the value.

Q Solve RR $a_n - 5a_{n-1} + 6a_{n-2} = 5^n$

Solⁿ $a_n - 5a_{n-1} + 6a_{n-2} = 5^n \quad \text{--- (1)}$

Second order RR.

$$m^2 - 5m + 6 = 0$$

$$(m-2)(m-3)$$

$$m = 2, 3 \quad \{ \text{distinct} \}$$

Homogenous solⁿ is

$$a_n^{(h)} = C_1(2)^n + C_2(3)^n$$

Since $f(n) = 5^n$ and 5 is not root of homogenous solⁿ

$$\text{Trial solⁿ} \Rightarrow a_n = A \cdot b^n$$

↓

$$a_n^{(p)} = A(5)^n \quad \text{--- (2)}$$

$$a_{n-1} = A 5^{n-1}$$

$$a_{n-2} = A \cdot 5^{n-2}$$

Putting the value of a_n in eqⁿ ①

$$A \cdot 5^n - 5(A \cdot 5^{n-1}) + 6(A \cdot 5^{n-2}) = 5^n$$

$$A \cdot 5^n - 5(A \cdot \frac{5^n}{5}) + 6(A \cdot \frac{5^n}{5^2}) = 5^n$$

$$A \cdot 5^n \left[1 - \frac{5}{5} + \frac{6}{25} \right] = 5^n$$

$$A \cdot 5^n \left[1 - 1 + \frac{6}{25} \right] = 5^n$$

$$\boxed{A = \frac{25}{6}}$$

Put $A = \frac{25}{6}$ in eq 2

$$a_n^{(p)} = \frac{25}{6} \cdot 5^n$$

$$= \frac{1}{6} \cdot 5^2 \cdot 5^n$$

$$\boxed{a_n^{(p)} = \frac{1}{6} \cdot 5^{n+2}}$$

Total solution of eqⁿ ① is

$$a_n = a_n^{(h)} + a_n^{(p)}$$

$$\boxed{a_n = c_1(2)^n + c_2(3)^n + \frac{1}{6} \cdot 5^{n+2}}$$

Q Solve recurrence relation $a_n - 7a_{n-1} + 10a_{n-2} = 3^r$
given that $a_0 = 0$ & $a_1 = 1$

Solⁿ $a_n - 7a_{n-1} + 10a_{n-2} = 3^r$ — (1)

characteristic eqⁿ is

$$m^2 - 7m + 10 = 0$$

$$(m-2)(m-5) = 0$$

$$m = 2, 5$$

Homogeneous solⁿ is

$$a_r(h) = c_1(2)^r + c_2(5)^r$$

Since $f(n) = 3^r$ & 3^r is not a root of homogeneous solution
will assume particular solution

$$a_n(p) = A \cdot (3)^r \quad \text{--- (2)}$$

$\therefore a_{r-1} = A \cdot (3)^{r-1}$

$$a_{r-2} = A \cdot (3)^{r-2}$$

Putting values in eqⁿ (1)

$$A \cdot 3^r - 7(A \cdot 3^{r-1}) + 10(A \cdot 3^{r-2}) = 3^r$$

$$A \cdot 3^r \left[1 - \frac{7}{3} + \frac{10}{9} \right] = 3^r$$

$$A = -\frac{9}{2}$$

Put $A = -\frac{9}{2}$ in eqⁿ (2)

$$a_4^{(p)} = -\frac{9}{2} 3^4$$

$$a_r^{(p)} = -\frac{1}{2} \cdot 3^{r+2}$$

Total Solⁿ

$$a_r = a_r^{(h)} + a_r^{(p)}$$

$$a_r = C_1(2)^r + C_2(5)^r - \frac{1}{2} \cdot 3^{r+2} \quad \text{--- (3)}$$

Putting $a_0 = 0$ i.e. $a_r = 0$ & $r = 0$ in eqⁿ (3)

$$C_1(2)^0 + C_2(5)^0 - \frac{1}{2} \cdot 3^2 = 0$$

$$C_1 + C_2 = \frac{9}{2} \quad \text{--- (4)}$$

Putting $a_1 = 1$ i.e. $a_r = 1$ & $r = 1$ in eqⁿ (3)

$$C_1(2) + C_2(5) - \frac{1}{2} \cdot 3^3 = 1$$

$$2C_1 + 5C_2 = \frac{27}{2} \quad \text{--- (5)}$$

Evaluate & find C_1 & C_2

$$2C_1 + 5C_2 = \frac{27}{2}$$

$$2C_1 + 2C_2 = 9$$

$$3C_2 = \frac{27}{2} - \frac{9}{1}$$

$$3C_2 = \frac{27-18}{2}$$

$$C_2 = \frac{9 \times 3}{2 \times 3}$$

$$C_2 = \frac{3}{2}$$

$$2C_1 + 5 \cdot \frac{3}{2} = \frac{27}{2}$$

$$2C_1 + \frac{15}{2} = \frac{27}{2}$$

$$\frac{4C_1 + 15}{2} = \frac{27}{2}$$

$$4C_1 = 27 - 15$$

$$4C_1 = 12$$

$$C_1 = 3$$

Putting the values of c_1 & c_2 in (3)

$$a_r = 3(2)^r + \frac{3}{2}(5)^r - \frac{1}{2}3^{r+2}$$

Case II when $f(x) = b^x$ & b is a root of characteristic eqn

1) If multiplicity of b is (1) i.e. b is a root of multiplicity 1 then

Suppose trial solⁿ is $a_r = A \cdot r b^r$

2) If multiplicity of b is (2) i.e. b is a root of multiplicity 2 then

Suppose trial solⁿ is $a_r = A \cdot r^2 b^r$

3) If multiplicity of b is 3,

Suppose trial solⁿ is $a_r = A \cdot r^3 b^r$

Q Solve recurrence relation $a_r - 3a_{r-1} + 2a_{r-2} = 2^r$

Solⁿ $a_r - 3a_{r-1} + 2a_{r-2} = 2^r$ — (1)

Characteristic eqⁿ is

$$m^2 - 3m + 2 = 0$$

$$m = 1, 2$$

Homogenous Solⁿ

$$a_r(h) = c_1(1)^r + c_2(2)^r$$

Since $f(n) = 2^r$ & 2 is characteristic root of multiplicity is 1

then particular Solⁿ is

$$a_r(p) = A \cdot n \cdot 2^r \quad \text{--- (2)} \quad \begin{aligned} a_{r-1} &= A \cdot (r-1) 2^{r-1} \\ a_{r-2} &= A \cdot (r-2) 2^{r-2} \end{aligned}$$

Putting values of a_r in eqⁿ (1)

$$A \cdot n \cdot 2^r - 3[A \cdot (n-1) 2^{r-1}] + 2[A \cdot (r-2) 2^{r-2}] = 2^r$$

$$A \cdot n \cdot 2^r - 3[A \cdot n \cdot 2^{r-1} - A \cdot 2^{r-1}] + 2[A \cdot n \cdot 2^{r-2} - 2A \cdot 2^{r-2}] = 2^r$$

$$A \cdot n \cdot 2^r - 3\left[A \cdot n \cdot \frac{2^r}{2} - A \cdot \frac{2^r}{2}\right] + 2\left[A \cdot n \cdot \frac{2^r}{2^2} - 2 \cdot A \cdot \frac{2^r}{2^2}\right] = 2^r$$

$$A \cdot n \cdot 2^r - \left[\frac{3A \cdot n \cdot 2^r}{2} + \frac{3A \cdot 2^r}{2} + \frac{2 \cdot A \cdot n \cdot 2^r}{4} - \frac{4 \cdot A \cdot 2^r}{4} \right] = 2^r$$

$$A \cdot n \cdot 2^r - \frac{3}{2} A \cdot n \cdot 2^r + \frac{3}{2} A \cdot 2^r + \frac{1}{2} A \cdot n \cdot 2^r - A \cdot 2^r = 2^r$$

$$A \cdot n \cdot 2^r + \frac{1}{2} A \cdot n \cdot 2^r - \frac{3}{2} A \cdot n \cdot 2^r + \frac{3}{2} A \cdot 2^r - A \cdot 2^r = 2^r$$

$$\frac{3}{2} A \cdot n \cdot 2^r - \frac{3}{2} A \cdot n \cdot 2^r + \frac{3}{2} A \cdot 2^r - A \cdot 2^r = 2^r$$

$$\frac{1}{2} A \cdot 2^r = 2^r \Rightarrow \boxed{A = 2}$$

Put $A = 2$ in eq (2)

$$a_r^{(P)} = 2 \cdot r(2)^r \\ = r \cdot 2^{r+1}$$

Total soln of eqn (1)

$$a_r = a_r^{(h)} + a_r^{(P)}$$

$$a_r = C_1(1)^r + C_2(2)^r + r \cdot 2^{r+1} \quad \underline{\text{Ans}}$$

Q Solve recurrence relation $a_n - 4a_{n-1} + 4a_{n-2} = 2^n$